Tom Paliotta

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EDUCATION

Bachelor of Science, Mechanical Engineering, May 2028
The University of Texas at Austin
Overall GPA: 3.91
Relevant Coursework: Statics, Dynamics, Solids, Fluids, Thermodynamics, Engineering Physics I–II, MATLAB
Our Lady of Good Counsel High School, May 2024
Overall GPA: 4.86
Top 5% of Graduating Class, Bill Jones STEM Award, State of Maryland Certificate of Merit, Student athlete

WORK EXPERIENCE

Intern, Parsons, 06/2025 – 08/2025
- Engineered optimized payload enclosures for unmanned aircraft, improving structural integrity, accessibility, and system reliability through iterative CAD modeling in SolidWorks
- Restored functionality to obsolete manufacturing equipment by designing and machining components on a 5-axis CNC, extending operational life and ensuring continued production capability
- Redesigned and fabricated high-efficiency resin casting molds, reducing cycle time and improving output consistency in the packaging of sensitive electronics

Intern, Johns Hopkins Applied Physics Laboratory, 06/2023 – 08/2023
- Utilized Fusion 360 to design and manufacture efficient heat management solutions in support of a naval power distribution system
- Selected to showcase final project design and analysis to 100+ conference attendees
- Worked with a mentor to learn the foundations of Power Engineering, manufacturing a bespoke charging pack for custom mobile use cases

Automotive Technician, Jiffy Lube, 05/2024 – 08/2024
- Performed maintenance and repairs on over 300 vehicles while collaborating with customers and team members to ensure efficient, high-quality service in a fast-paced environment
- Cultivated expertise in automotive systems including suspension, control and steering, power distribution, and load distribution

RESEARCH EXPERIENCE

Engine and Combustion Systems, University of Texas at Austin, 01/2026 – Present
- Investigating spark plug flashover behavior under varying combustion pressures, electrode gaps, and insulator configurations
- Supporting experimental work on rotating-cylinder diesel engine concepts to reduce friction losses, including testing under boosted (supercharged) operating conditions

SKILLS

- CAD Software: SolidWorks, Fusion 360, Autodesk CFD, SolidWorks FEA, KiCad, Autodesk Inventor, OnShape
- Engineering Tools: 5-axis CNC Machining, 3D Printing, Manual Milling, Vacuum Casting, PCB Milling, Soldering